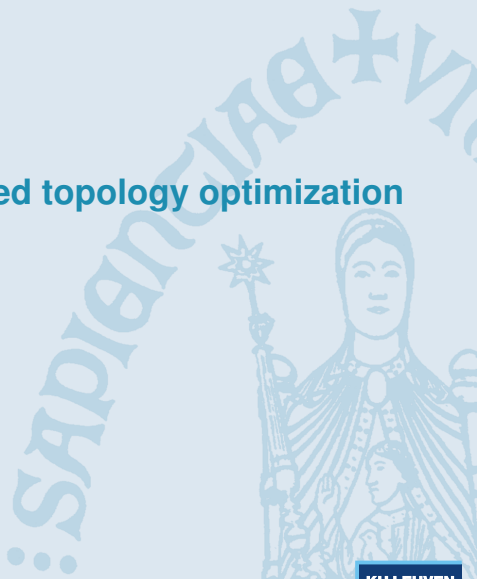
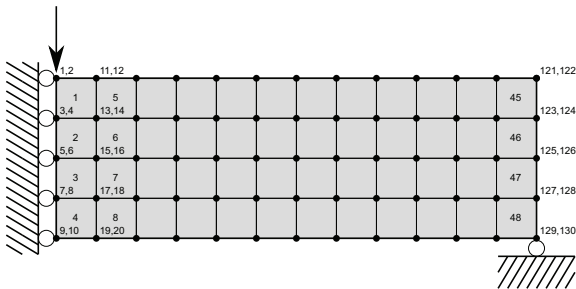


Chapter 9 – Density-based topology optimization



Example: MBB beam* (Messerschmitt-Bölkow-Blohm)



- The aim is to distribute a given amount $V_{\max}$ of material in the design domain such that the compliance of the structure is minimized.
- The design domain is divided into 300×100 four-node plane-stress finite elements.
- Each element e is assigned a density ρ_e representing the presence or absence of material: $\rho_e = 0$ corresponds to void, $\rho_e = 1$ corresponds to solid.
- Intermediate densities are also allowed in order to obtain a continuous optimization problem.

*N. Olhoff, M.P. Bendsøe, and J. Rasmussen. On CAD-integrated structural topology and design optimization. CMAME, 1991.

Formulation of the optimization problem

$$\begin{aligned} \min_{\boldsymbol{\rho}} : \quad & C = \mathbf{F}^T \mathbf{u} \\ \text{s.t.} : \quad & V = \sum_{e=1}^n v_e \rho_e \leq V_{\max} \\ & \mathbf{0} \leq \boldsymbol{\rho} \leq \mathbf{1} \end{aligned}$$

- The density ρ_e controls the Young's E_e modulus of element e . For now we adopt a linear relation between ρ_e and E_e :

$$E_e = E_{\min} + \rho_e(E_0 - E_{\min})$$

where E_0 and $E_{\min}$ are the Young's moduli of the solid and void phases, respectively.

- The displacements $\mathbf{u}$ are obtained as the solution of the state problem:

$$\mathbf{K} \mathbf{u} = \mathbf{F}$$

- The global stiffness matrix $\mathbf{K}$ is obtained as:

$$\mathbf{K} = \sum_{e=1}^n E_e \mathbf{K}_e^0 = \sum_{e=1}^n (E_{\min} + \rho_e(E_0 - E_{\min})) \mathbf{K}_e^0$$

where $\mathbf{K}_e^0$ is the element stiffness matrix of element e with a unit Young's modulus.

Sensitivities

- The sensitivity of the compliance C with respect to the element density ρ_e is obtained by means of the adjoint method:

$$\frac{\partial C}{\partial \rho_e} = \boldsymbol{\lambda}^T \left(\cancel{\frac{\partial \mathbf{F}}{\partial \rho_e}} - \frac{\partial \mathbf{K}}{\partial \rho_e} \mathbf{u} \right) = -\frac{\partial E_e}{\partial \rho_e} \boldsymbol{\lambda}^T \mathbf{K}_e^0 \mathbf{u} = -(E_0 - E_{\min}) \boldsymbol{\lambda}^T \mathbf{K}_e^0 \mathbf{u}$$

where the vector $\boldsymbol{\lambda}$ is found as the solution of the adjoint problem:

$$\mathbf{K} \boldsymbol{\lambda} = \mathbf{F}_a$$

For minimum compliance problems, the adjoint problem is identical to the state problem, which implies that $\boldsymbol{\lambda} = \mathbf{u}$, or:

$$\frac{\partial C}{\partial \rho_e} = -(E_0 - E_{\min}) \mathbf{u}^T \mathbf{K}_e^0 \mathbf{u}$$

- The sensitivity of the volume V with respect to the element density ρ_e is given by:

$$\frac{\partial V}{\partial \rho_e} = v_e$$

Optimality criteria method

- An increase of the density ρ_e of element e always leads to a lower compliance C and a higher volume V .
- The sensitivity ratio $-\frac{\partial C/\partial \rho_e}{\partial V/\partial \rho_e}$ is a measure for the material efficiency of element e : it expresses to which extent the compliance is reduced by investing a volume unit of material in element e .
- We can improve the design by redistributing material in such a way that the investment in material efficient elements is increased and the investment in material inefficient materials is reduced (without violating the box constraints). This is the basis of the optimality criteria method:

$$\rho_e^{\text{new}} = \max(0, \max(\rho_e - m, \min(1, \min(\rho_e + m, \rho_e B_e^\alpha))))$$

where $m = 0.2$ is a move limit, $\alpha = 0.5$ is a damping factor, and B_e is calculated as:

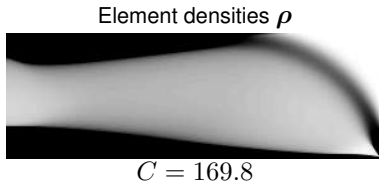
$$B_e = -\frac{1}{\lambda} \frac{\partial C/\partial \rho_e}{\partial V/\partial \rho_e}$$

The factor λ must be determined iteratively in such a way that after updating, the volume constraint is exactly satisfied.

Optimized design

The optimality criteria method is used to update the element densities ρ .

→ The optimized design contains a lot of gray elements.



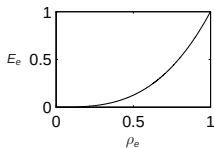
Penalization

The SIMP method (Solid Isotropic Material with Penalization) is used to penalize intermediate densities:

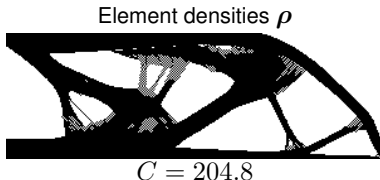
$$E_e = E_{\min} + (E_0 - E_{\min})\rho_e^p \quad \text{where } p = 3$$

Impact on sensitivities:

$$\frac{\partial C}{\partial \rho_e} = -p(E_0 - E_{\min})\rho_e^{p-1} \lambda^T \mathbf{K}_e^0 \mathbf{u}$$



- Gray elements are effectively suppressed.
- The solution is mesh dependent.
- Checkerboard patterns occur.
- No length scale control.

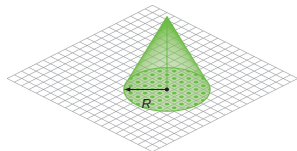


Sensitivity filtering

The sensitivities $\partial C / \partial \rho_e$ are averaged over a circular filter region with radius R :

$$\widehat{\frac{\partial C}{\partial \rho_e}} = \frac{1}{\max(\gamma, \rho_e)} \frac{\sum_{i=1}^n H_{ei} \rho_i \frac{\partial C}{\partial \rho_i}}{\sum_{i=1}^n H_{ei}}$$

where $H_{ei} = \max(0, R - r_{ei})$ and $\gamma = 10^{-3}$.



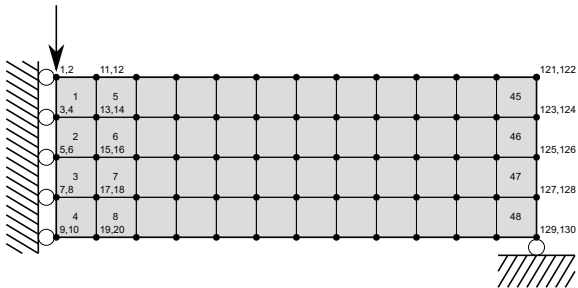
- Mesh independent solution.
- No checkerboard patterns.
- Length scale control.
- Gray elements.

Element densities ρ



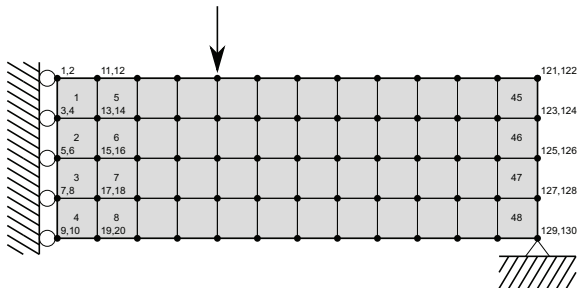
$C = 222.3$

MATLAB example



- Solve the optimization problem using sensitivity filtering (ex1a.m and ex1b.m).
- Test the influence of the volume fraction, the penalization power, the filter radius, the discretization (number of finite elements), and the direction of the force (downwards or upwards) on the optimized design.

MATLAB example



- Change the loading and boundary conditions as specified on the figure. Solve the optimization problem with a volume fraction of 0.2 (ex2.m).

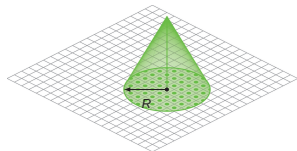
Density filtering

Element densities are averaged over a circular filter region with radius R :

$$\tilde{\rho}_e = \frac{\sum_{i=1}^n H_{ei} \rho_i}{\sum_{i=1}^n H_{ei}}$$

Impact on sensitivities (where ψ can be C or V):

$$\frac{\partial \psi}{\partial \rho_j} = \sum_{e=1}^n \frac{\partial \psi}{\partial \tilde{\rho}_e} \frac{\partial \tilde{\rho}_e}{\partial \rho_j} = \sum_{e=1}^n \frac{\partial \psi}{\partial \tilde{\rho}_e} \frac{H_{ej}}{\sum_{i=1}^n H_{ei}}$$



Design variables ρ

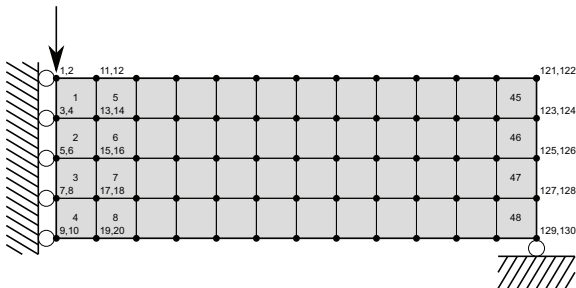


Element densities $\tilde{\rho}(\rho)$



$$C = 238.3$$

MATLAB example

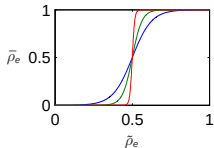


- Solve the optimization problem using density filtering (ex1a.m and ex1b.m).
- Change the MATLAB code to use MMA instead of the optimality criteria method (ex3a.m).
- Change the MATLAB code to use GCMMA (ex3b.m).

Heaviside projection

Light gray elements are projected to white; dark gray elements are projected to black:

$$\bar{\rho}_e = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\rho}_e - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}$$



Impact on sensitivities (where ψ can be C or V):

$$\frac{\partial \psi}{\partial \rho_j} = \sum_{e=1}^n \frac{\partial \psi}{\partial \bar{\rho}_e} \frac{\partial \bar{\rho}_e}{\partial \tilde{\rho}_e} \frac{\partial \tilde{\rho}_e}{\partial \rho_j} \quad \text{where} \quad \frac{\partial \bar{\rho}_e}{\partial \tilde{\rho}_e} = \frac{\beta \operatorname{sech}(\beta(\tilde{\rho}_e - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}$$

Design variables ρ



Filtered variables $\tilde{\rho}(\rho)$

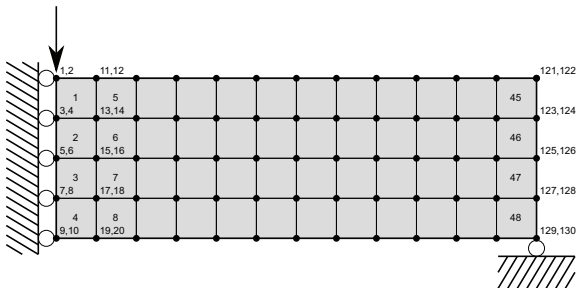


Element densities $\bar{\rho}(\tilde{\rho}|\eta)$



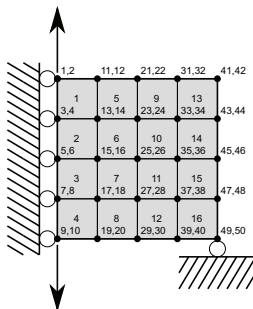
$$C = 193.7$$

MATLAB example



- Solve the optimization problem using the Heaviside projection filter in combination with MMA (`ex4.m`).
- Test the influence of the projection threshold η on the optimized design: solve the optimization problem with $\eta = 0$, $\eta = 0.5$, and $\eta = 1$.

Multiple load cases



- Solve the problem considering a single load case consisting of the two point loads (ex5a.m).
- Solve the problem by minimizing the sum of the compliance for two load cases each consisting of a single point load (ex5b.m).
- Solve the problem by minimizing the maximum compliance for each of the two load cases (ex5c.m).

Min-max problems

A min-max problem is formulated as follows:

$$\begin{aligned} \min_{\mathbf{x}} : \quad & \max_{i=1,\dots,p} \{h_i(\mathbf{x})\} \\ \text{s.t.} : \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, q \\ & \mathbf{x} \in \mathcal{X} \end{aligned}$$

Assuming that all functions h_i are positive, this problem can be reformulated as:

$$\begin{aligned} \min_{\mathbf{x}, z} : \quad & z \\ \text{s.t.} : \quad & h_i(\mathbf{x}) \leq z, \quad i = 1, \dots, p \\ & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, q \\ & \mathbf{x} \in \mathcal{X}, \quad z \geq 0 \end{aligned}$$

This problem can be solved using MMA with the following parameters (see also `help mma`):

```
f0 = 0;  
f = [h; g];  
a0 = 1;  
a = [ones(p,1); zeros(q,1)];  
c = 10000;  
d = 0;
```